



Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
MCA Department

PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Class: SYMCA	Div: B	Course Code: MCA01604	Batch: S2
Semester: III		Course Name: Data Science Laboratory	
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CO No: CO605.2		Assignment No: 09	

Title:

Use the FP-Growth Algorithm to perform Market Basket Analysis on a retail dataset. You will extract frequent item sets and generate association rules to discover purchasing patterns and help improve sales strategies

Code:

```
import pandas as pd
from mlxtend.frequent_patterns import fpgrowth, association_rules
from mlxtend.preprocessing import TransactionEncoder

try:
    df = pd.read_csv('groceries.csv').drop(columns=['Item(s)'])
    print("□ Dataset loaded successfully!")
except FileNotFoundError:
    print("□ Error: 'groceries.csv' not found. Please save the data to a file first.")
    exit()

transactions = []
for index, row in df.iterrows():
    basket = row.dropna().tolist()
    transactions.append(basket)

print(f"\nSuccessfully created {len(transactions)} transaction baskets.")
print("Example basket:", transactions[0])

te = TransactionEncoder()
te_ary = te.fit(transactions).transform(transactions)
df_encoded = pd.DataFrame(te_ary, columns=te.columns_)

frequent_itemsets = fpgrowth(df_encoded, min_support=0.005, use_colnames=True)
```



```
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.1)

print("\n--- Top 10 Association Rules (sorted by Lift) ---")
if rules.empty():
    print("No association rules found. Try lowering the 'min_support' or 'min_confidence' values.")
else:
    rules_sorted = rules.sort_values(by='lift', ascending=False)
    print(rules_sorted.head(10)[['antecedents', 'consequents', 'support', 'confidence', 'lift']])
```

Output:

```
✓ Dataset loaded successfully!

Successfully created 9835 transaction baskets.
Example basket: ['citrus fruit', 'semi-finished bread', 'margarine', 'ready soups']

--- Top 10 Association Rules (sorted by Lift) ---
      antecedents  ...  lift
262  (whole milk, tropical fruit)  ...  5.212371
259    (yogurt, root vegetables)  ...  5.212371
258    (yogurt, whole milk)  ...  4.828814
263  (root vegetables, tropical fruit)  ...  4.828814
384    (pip fruit, root vegetables)  ...  4.716272
1634                (ham)  ...  4.639851
1635    (white bread)  ...  4.639851
253  (root vegetables, tropical fruit)  ...  4.454257
959    (yogurt, other vegetables)  ...  4.395570
960  (whole milk, fruit/vegetable juice)  ...  4.395570

[10 rows x 5 columns]
```